

Workforce Continuity Agent

AI-Powered Automatic Task Reassignment

Team: Ascent Hack

Date: May 2026

The Problem

When an employee is absent unexpectedly, their pending tasks stall. Managers must manually triage each task: find a replacement, check skills, rebalance workloads. This process is slow, error-prone, and creates bottlenecks that ripple across teams.

Current solutions are reactive and manual. There is no intelligent system that automatically evaluates task complexity, matches skills across the team, and either reassigns tasks to the best available person or completes them autonomously using AI.

Our Solution

The Workforce Continuity Agent is a multi-agent AI system that detects employee absence and automatically handles all pending tasks through a 6-phase pipeline. It analyzes tasks, evaluates team capacity, makes reallocation decisions, and can even complete tasks autonomously using AI code generation and web search.

6-Phase Pipeline

- Phase 1 - Detection: Absence detected via webhook or manual trigger
- Phase 2 - Retrieval: Pending tasks retrieved, GitHub repo synced
- Phase 3 - Analysis: Tasks analyzed for skills, complexity, estimated hours
- Phase 4 - Reallocation: Team scored, tasks reassigned or marked for AI
- Phase 5 - Execution: AI completes tasks (web search, code gen, git push)
- Phase 6 - Reporting: Manager report with decision rationale generated

Agent Architecture

The system uses 5 specialized agents coordinated by a central orchestrator. Each agent has a distinct responsibility and they communicate through a shared activity bus with WebSocket broadcasting.

Agent	Responsibility
Orchestrator	Central pipeline controller driving all 6 phases
Task Analyzer	LLM analysis: skills, complexity, task splitting
Availability Agent	Skill matching and workload scoring
Reallocation Agent	Threshold decisions: reassign or auto-complete
Executor Agent	Code gen, web search, file write, git push

Technology Stack

Component	Technology
Backend	FastAPI (Python)
Frontend	React + Vite + Tailwind CSS
AI Engine	OpenRouter (Gemini 2.0 Flash)
Database	SQLite (file-based)
Version Control	GitHub API + Git CLI
Web Search	DuckDuckGo API

Key Design Decisions

- Hybrid autonomy: reassign when confidence > 0.6, otherwise AI completes
- Graceful degradation: LLM failures fall back to heuristics
- Real side effects: actual git commits, DB updates, notifications
- Async pipeline: HTTP returns immediately, pipeline runs in background

Autonomy in Practice

The system operates on a spectrum of autonomy. For each task, the agent evaluates whether a human teammate can handle it better or if AI should complete it autonomously.

Real-World Scenario

A frontend developer (P5) is absent with 3 pending tasks:

- Cart System (CODE, 6h): P1 has 90% React match, score 0.78. Reassigned.
- API Integration (CODE, 4h): No frontend dev has backend skills. Auto-completed by AI.
- Design Review (REVIEW, 2h): P6 has relevant skills, score 0.82. Reassigned.

Verifiable Side Effects

- Database: Task statuses updated, assignments changed
- GitHub: Real commits pushed with descriptive messages
- Notifications: Team members notified of assignments
- Reports: Manager reports with full decision rationale
- WebSocket: Complete activity log of every agent action

Impact Metrics

Metric	Manual	With Agent
Triage 10 tasks	2-4 hours	Under 2 minutes
Skill matching	Judgment	Data-driven
Tasks missed	Common	Zero
After-hours	None	24/7 autonomous

Conclusion

The Workforce Continuity Agent demonstrates that multi-agent AI systems can meaningfully reduce organizational friction caused by employee absence. By combining intelligent task analysis, skill-based reallocation, and autonomous execution, it ensures work continues regardless of team disruptions.